



# DAVID JÖNSSON

DATA SCIENTIST & ENGINEER

## SUMMARY

**Data Scientist and Engineer** specializing in the application of advanced Deep Learning architectures to complex physical systems. Currently **completing a 5-year M.Sc. in 4 years** (Accelerated Track), demonstrating exceptional time management and rapid technical acquisition. My Master's research at the **Swedish Defence Research Agency (FOI)** applies **Transformer models** to chemical structure prediction, which will be **published later this year**. I bring a unique "Northern" work ethic - pragmatic, resilient, and collaborative - combined with a legal background that ensures strict adherence to compliance and ethical standards in sensitive data environments.

## CONTACT

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## PERSONAL DATA

- **Date of Birth:** 1999-12-08
- **Nationality:** Sweden
- **Languages:** Swedish (Native), English (Professional)
- **Security Cleared:** i.e. no criminal records/drugs

## EDUCATION

### Master's Thesis | Umeå University, Sweden

- **Transformer Architecture Implementation:** Developing "SpecTUS," a Transformer-based encoder-decoder model that treats mass spectral data as a sequence-to-sequence translation task.
- **Fine-Tuning & Data Scarcity:** Implementing a "forward-inverse loop" strategy to pre-train models on synthetic data before fine-tuning on experimentally curated datasets, addressing data scarcity and the "long tail" problem.
- **Complex Data Curation:** Curating and preprocessing a dataset of 300,000 samples (NIST), utilizing tokenization methods (SELFIES) to prevent invalid syntax generation in machine learning outputs.

### M.Sc. Industrial Engineering & Management | Umeå University, Sweden

2022/08 – 2026/06 (Accelerated Track)

- Accelerated track: **The only person out of 70 people to complete my degree at faster speed** - in 4 years instead of 5.
- Specialization: Data Science, Machine Learning and Financial Management

### Law Programme (Juristprogrammet) | Umeå University, Sweden

08/2020 – 01/2021

- Scope: 105 ECTS (approx. 2 years full-time).
- Relevance: Gained deep understanding of Civil Law, EU Regulations, and the legal workflow.

## RELEVANT WORK EXPERIENCE

### Teaching Assistant (Python) | Umeå University

08/2025 – 2026-01-01

- Guiding first-year engineering students through **Python programming** fundamentals.

### Business Intelligence Intern | BAE Systems

Häggglunds, Sweden, 06/2025 – 08/2025

- Investigated the utility of **LLM AI Agents**, its potential and created a 10 page document around how to implement it in the defense sector.
- Held lectures for the procurement department (120 people) about effective **LLM Prompt Engineering** and built **Power BI** solutions to visualize complex workflow bottlenecks.

## OTHER EXPERIENCES

### Sports event Coordinator | Umeå University

08/2022 – 06/2026

- Planned Sports-related events for the institution

### Project Manager: Lawyers Day | Law Student Association at Umeå University, 2021

- I shared overall responsibility with two others for Juristens Dag, the largest career fair for the Law Program at Umeå University. We managed a budget of nearly half a million SEK and led a team of 14 people to ensure the success of the event, which hosted around 40 participating companies.

### Young Entrepreneurs Nordic Meetup, 2017

- Selected as one of ten individuals under the age of 36 to represent Sweden at this networking event in Helsinki.

### Scholarship | Indivator, 2017

- Scholarship worth 10 000 Sek for a entrepreneurship course.